

Rendering of Synthetic Underwater Images Towards Restoration

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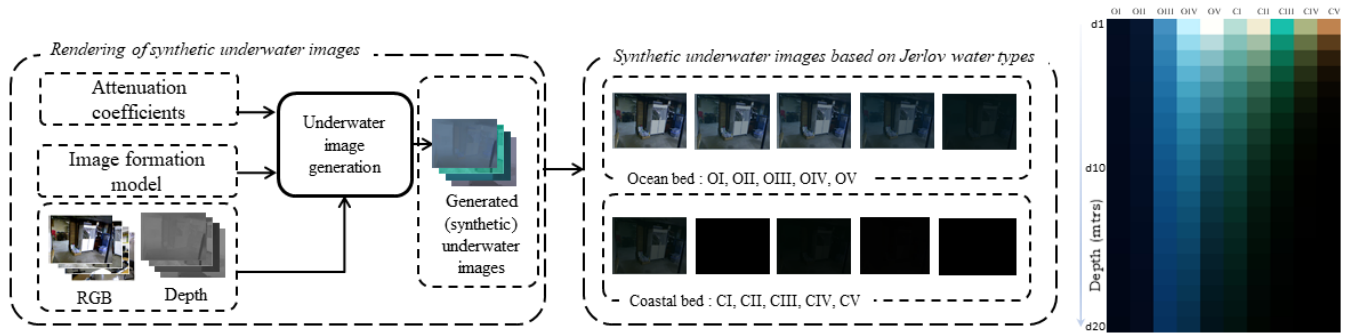


Figure 1: Rendering of synthetic underwater images based on Jerlov [Tyler 1968] water types.

ABSTRACT

In this paper, we propose to render synthetic underwater images considering revised image formation model, towards modeling restoration. Underwater images suffer from low contrast, color cast and haze due to dynamically varying properties of water, floating particles and submerged sediments. Due to this, light reaching the object, from the surface is subjected to direct scattering, backscattering and forward scattering. We propose to model this varying nature of light to render synthetic underwater images based on depth information and inherent optical properties of Jerlov water types.

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1 INTRODUCTION

In this work, we render synthetic underwater images based on revised image formation model for learning restoration. Revised image formation model is sensitive to inherent optical properties

(IOP) of Jerlov water types. These IOP's are sensitive to visible spectrum (400nm to 700nm) and needs to be modeled accordingly. However, traditional methods [Li et al. 2017] assumed these attenuation coefficients to be spectrally uniform and ignored the sensitivity. Towards this, we render synthetic underwater images considering inherent optical properties of Jerlov water types based on visible spectrum. Rendered synthetic underwater images can be used to train complex architectures for modeling restoration.

Restoration of images involves removing of blur, noise and other degradations that occurs during the capture, with the aim of producing exact representations of the scene. Applications of restoration methods are spread across wide areas such as, object detection, medical image analysis, precision agriculture and marine archaeology etc. Although, image restoration is a mature field of research, significant progress in this field is continuously being made. Recently image restoration has emerged as an active research area in underwater applications. Researchers aspire to capture high quality underwater images towards inspections of marine archaeology, coral-reefs, aquatic life, flora and fauna. Several authors propose learning and non learning based methods towards modeling restoration.

Major challenges in learning-based architectures is the need for a larger database coupled with labels or corresponding ground truth information. Generating ground truth labels in underwater scenarios demands in-situ measurements using expensive equipment [1986]. Performing in-situ measurements requires skilled personnel to handle complex equipment and is a challenging affair. Gathering larger chunks of data with corresponding depth and ground truth information in the underwater scenario is challenging and remains an open problem. Towards this, we propose to render synthetic

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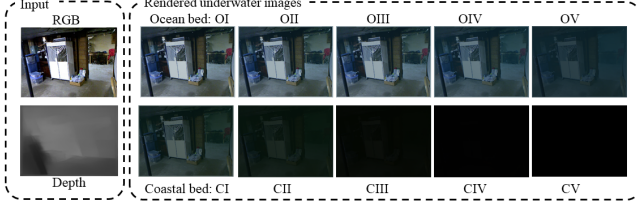


Figure 2: Rendered synthetic underwater images for ocean and coastal bed.

underwater images considering revised image formation model [2018] as shown in Figure 1. We validate rendered synthetic images by training conditional generative adversarial networks (CGAN) towards modeling restoration. In addition, we test the trained model on benchmark underwater dataset [2020].

2 RENDERING OF SYNTHETIC UNDERWATER IMAGES

We propose to render synthetic underwater images using revised image formation model as shown in Equation 1. Revised image formation model is sensitive to S_c , E , b and β coefficients. We estimate these coefficients as given by authors in [2018]. Estimated B_c^∞ accordingly is depicted in Figure 1 on right side. Rendered synthetic underwater images are shown in Figure 2 and are useful in training deep neural networks towards modeling restoration.

$$I(x) = J(x)e^{-\beta_c^D(v_D)d(x)} + B_c^\infty(1 - e^{\beta_c^B(v_B)d(x)}) \quad (1)$$

where $v_D = \{d(x), \rho, E, S_c, \beta\}$ and $v_B = \{E, S_c, \beta, b\}$. Here $d(x)$ is the depth range, ρ is the reflectance of each pixel in the scene, S_c is the spectral response of the camera, E is the spectral irradiance of the scene, b is scattering coefficient and β is beam attenuation coefficient.

3 RESTORATION USING RENDERED SYNTHETIC UNDERWATER IMAGES

We model conditional generative networks (CGAN) to learn mapping from rendered synthetic underwater image to ground truth image. Rendered synthetic underwater images facilitate training deep neural networks effectively and is evident in Figure 3 and 4. We prove quantitatively proposed methodology outperforms with reference to state of art methods.

4 RESULTS AND DISCUSSIONS

In this section, we present results of restoration trained using rendered synthetic underwater images. We evaluate the proposed methodology using appropriate quantitative metrics and compare the same with state-of-the-art techniques. Figure 3 shows restoration results of synthetic underwater images. Figure 4 shows restoration results of real underwater images with the corresponding quantitative scores.



Figure 3: Row I: Rendered synthetic underwater images, Row II: Restoration results of our methodology.

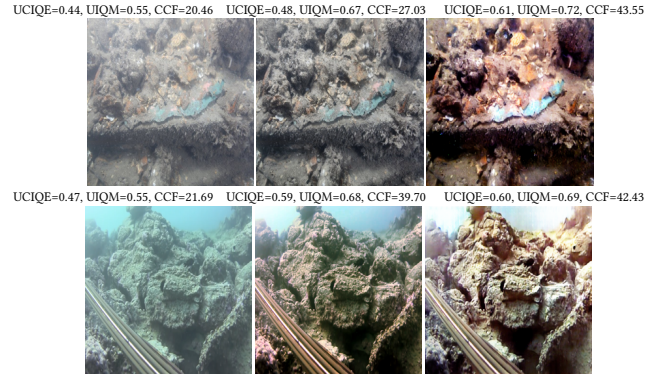


Figure 4: Column I: Input images, Column II: Restoration results of authors in [2020], Column III: Restoration results of our methodology.

5 CONCLUSIONS

In this work, we have proposed to render synthetic underwater images considering revised image formation model towards learning restoration. We have modeled varying nature of light to render synthetic underwater images based on depth information and inherent optical properties of Jerlov water types. Results of restoration as shown in Figure 3 and 4 prove, rendered synthetic underwater images are realistic in nature and are inline with real underwater images.

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